

MUHAMMAD SHOAIB

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Software Engineering student at COMSATS University Lahore with hands-on experience in Java, SQL, and the MERN stack. Strong foundation in Data Structures and Algorithms, with a passion for building scalable and efficient web and mobile applications. Proven ability to develop full-stack solutions and deliver high-quality projects from concept to deployment.

PROJECTS & TECHNICAL EXPERIENCE

Employee Management System

Enterprise Employee Records Management | Java (OOP)

- Developed an Employee Management System using Java with object-oriented programming principles, implementing full CRUD operations for managing employee records.
- Utilized file handling techniques for data persistence and implemented data validation for error prevention.
- Applied inheritance, polymorphism, and encapsulation to create maintainable and scalable code architecture.
- Designed user-friendly console interface for efficient data entry and retrieval operations.

Advanced Chat Application

Algorithm-Optimized Messaging System | Java & Data Structures

- Designed and developed a chat application in Java by applying advanced Data Structures and Algorithms including hash maps, queues, and binary trees for optimized message handling.
- Implemented custom algorithms to reduce message retrieval time by 50% in multi-user environments.
- Applied object-oriented design patterns including Observer and Singleton for scalable architecture.
- Conducted unit testing and debugging to ensure 99% uptime and reliability during load testing.

Supermarket Management System

Retail Operations & Inventory Management Solution | SQL & C#

- Designed and implemented a relational database system for supermarket management, including inventory tracking, billing automation, and sales analytics using SQL Server.
- Developed a desktop application in C# with Windows Forms for user-friendly data entry and reporting.
- Created normalized database schemas with proper indexing, improving query performance by 40%.
- Implemented stored procedures and triggers for automated inventory updates and sales calculations.

Ride Booking Application

Complete UI/UX Design Prototype | Figma

- Created a comprehensive ride booking application prototype in Figma, including wireframes, user flows, and high-fidelity mockups for both passenger and driver interfaces.
- Conducted user research and usability testing to optimize booking flow, reducing steps by 25%.
- Developed design system with reusable components, ensuring consistency across 30+ screens.
- Focused on accessibility standards and mobile-first design principles for enhanced user experience.

Real-Time Chat Application

Cross-Platform Mobile Messaging App | Flutter & Firebase

- Built a cross-platform mobile chat application using Flutter and Firebase, featuring real-time messaging, user authentication, and cloud-based data synchronization.
- Integrated Firebase Firestore for instant message delivery and Firebase Authentication for secure user login.
- Designed intuitive UI/UX with Material Design principles, enhancing user engagement and app usability.
- Deployed application to both iOS and Android platforms, reaching 100+ test users during beta phase.

Mian Autos

Full-Stack Vehicle Marketplace Platform | MERN Stack

- Developed a full-stack vehicle listing and marketplace platform using MongoDB, Express.js, React, and Node.js, enabling users to browse, search, and list vehicles with dynamic filtering capabilities.
- Implemented RESTful APIs for seamless front-end and back-end communication, reducing data retrieval time by 30%.
- Built responsive UI components with React, ensuring optimal user experience across desktop and mobile devices.
- Designed and optimized database schemas in MongoDB for efficient data storage and retrieval of vehicle listings.

EDUCATION

COMSATS University Islamabad, Lahore Campus

Bachelor of Science in Software Engineering

Expected 2027

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, Web Technologies

Intermediate Education (FSc Pre-Engineering)

2020 – 2022

Matriculation (Secondary School Certificate)

2018 – 2020

CERTIFICATIONS

Flutter Development Certification

September 2025 – November 2025

Elevare Tech - Comprehensive training in cross-platform mobile development with Flutter framework

SKILLS & TECHNOLOGIES

Programming Languages: Java, Python, C++, SQL, JavaScript

Frameworks & Libraries: Flutter, React, Express.js, Node.js

Databases: MongoDB, SQL Server, Firebase Firestore

Tools & Platforms: Git, GitHub, Figma, VS Code, Android Studio

Core Competencies: Object-Oriented Programming (OOP), Data Structures & Algorithms, RESTful API Development, Database Design, Responsive Web Design, Mobile App Development, UI/UX Design, Agile Methodology